Automotive NTN UE Antenna Sizing Study

How large does a 3GPP n255 / n256 patch array become on a vehicle – and is that size realistic?

Closed-form sizing study (aperture, scan, link budget)
Bands: n255 L-band 1525–1559 MHz (Rx) / n256 S-band 2170–2200 MHz (Rx)
RO4003C · h = 3.05 / 1.52 mm
2026-11-08

Contents

1 Executive summary
2 Scope, method and assumptions
3 Bands and element
4 Array size versus gain
5 Scan and packaging
6 Link budget
7 Dual-band and duplex constraints
8 Recommendation
9 Limitations and next steps
10 Reproduction

1 Executive summary

This study answers a packaging question, not a full-wave question: how much roof area does an electronically steered patch array need for a 3GPP NTN user terminal in the n255 (L-band) and n256 (S-band) bands, and does that fit on a passenger car? The answer is obtained analytically from the lambda/2 lattice, the aperture directivity, a scan-loss model and a downlink link budget – no FDTD run is required to make the packaging decision.

- The element is large: 50.7 × 64.4 mm at n255 and 36.0 × 45.5 mm at n256, on a λ/2 lattice of 96.1 mm and 68.1 mm.
- A 2×2 array already needs 27 cm (n255) and 19 cm (n256) of module edge length; 4×4 needs 46 cm and 33 cm.
- n256 is feasible on a car roof: 2×2 fits a roof pod, 5×5 fits the roof centre. n255 is tight: 2×2 needs 27 cm, i.e. more than a 25 cm roof pod; 3×3 fits the roof centre.
- From the link budget, 2×2 closes a low-rate NB-NTN link with large margin; even the 85 dBHz broadband-class target is reached with 2×2 (n255) and 3×3 (n256) at the assumed EIRP. The packaging size, not the link, is the binding constraint.
- n255 and n256 cannot share one aperture: the n255 patch is larger than the n256 lattice cell. Two separate modules (or a broadband shared aperture) are required.

2 Scope, method and assumptions

All numbers are closed-form. The chain is: substrate and band → patch dimensions (Hammerstad) → λ/2 lattice at the upper band edge → aperture directivity $D = 4\pi A/\lambda^2$ → feed and scan losses → downlink C/N_0. The model is deliberately conservative where a full-wave result would be needed: the feed loss is a flat 1.5 dB and the scan loss adds a patch-like element roll-off on top of the aperture projection.

Table 1. Assumptions used throughout the study.

Item	Value	Comment
Substrate	Rogers RO4003C	$\varepsilon_r = 3.55$, $\tan\delta = 0.0027$
Lattice	λ/2 at band edge	no grating lobes up to 90°
Feed loss	1.5 dB	corporate feed + radome
Scan loss model	$15 \cdot \log10(\cos\theta_0)$	aperture + element roll-off
Satellite altitude	630 km	Amazon Leo upper shell
Satellite EIRP	30 dBW	placeholder – replace with operator data
Receiver NF	1.5 dB	automotive-grade LNA
Targets	55 / 70 / 85 dBHz	NB-NTN / moderate / broadband class

3 Bands and element

3GPP defines the NTN bands n255 (L-band, Rx 1525–1559 MHz, Tx 1626.5–1660.5 MHz) and n256 (S-band, Rx 2170–2200 MHz, Tx 1980–2010 MHz). The array is sized for the receive direction, which is the G/T-critical one; the transmit direction uses the same aperture with a diplexer.

Table 2. Band parameters and resulting element geometry.

Quantity	n255	n256
Receive band	1525–1559 MHz	2170–2200 MHz
Transmit band	1626.5–1660.5 MHz	1980–2010 MHz
Centre frequency	1.542 GHz	2.185 GHz
λ_0 at centre	194 mm	137 mm
Lattice $\lambda/2$ (band edge)	96.1 mm	68.1 mm
Substrate height	3.05 mm	1.52 mm
Patch length L	50.7 mm	36.0 mm
Patch width W	64.4 mm	45.5 mm

The element is a single-layer rectangular patch on RO4003C, dimensioned with the Hammerstad relations at the band centre. The lattice is set by the upper band edge so that no grating lobe can appear even at large scan angles. Figure 1 shows both elements to scale together with their lattice cells.

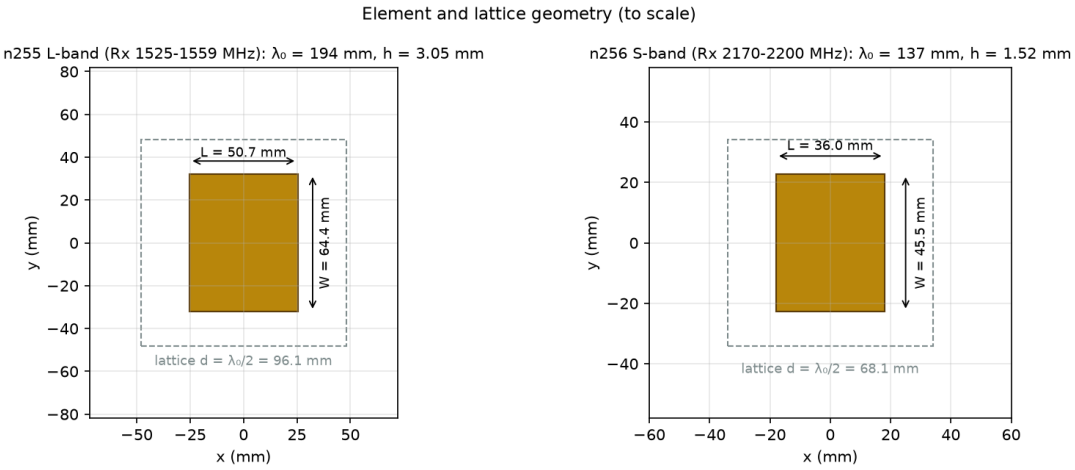


Figure 1. Element and lattice geometry at both bands (to scale).

4 Array size versus gain

For an N×N array on a $\lambda/2$ lattice the aperture is $(N\cdot\lambda/2)^2$, so $D = 4\pi A/\lambda^2$ collapses to $D = \pi\cdot N^2$ (linear), i.e. $10\cdot\log10(\pi N^2)$ dBi. Every doubling of N therefore adds 6.02 dB – independent of the band, because the lattice scales with the wavelength.

Table 3 lists aperture, module size, directivity, realised broadside gain (1.5 dB feed loss) and the gain at 60° scan for both bands. Figure 2 shows gain versus module edge length – the gain itself depends only on N, but the

same gain needs a smaller module at n256. Section 5 then compares those module sizes with the areas available on a vehicle (Figure 4).

Table 3. Array metrics for both bands (Rx).

N×N	Elements	Aperture	Module	D	G	G at 60°
n255 2×2	4	19 cm	27 cm	10.9 dBi	9.4 dBi	4.9 dBi
n255 3×3	9	29 cm	37 cm	14.4 dBi	12.9 dBi	8.4 dBi
n255 4×4	16	38 cm	46 cm	16.9 dBi	15.4 dBi	10.9 dBi
n255 5×5	25	48 cm	56 cm	18.9 dBi	17.4 dBi	12.8 dBi
n255 6×6	36	58 cm	65 cm	20.4 dBi	18.9 dBi	14.4 dBi
n255 8×8	64	77 cm	85 cm	22.9 dBi	21.4 dBi	16.9 dBi
n256 2×2	4	14 cm	19 cm	10.9 dBi	9.4 dBi	4.9 dBi
n256 3×3	9	20 cm	26 cm	14.5 dBi	13.0 dBi	8.4 dBi
n256 4×4	16	27 cm	33 cm	17.0 dBi	15.5 dBi	10.9 dBi
n256 5×5	25	34 cm	40 cm	18.9 dBi	17.4 dBi	12.9 dBi
n256 6×6	36	41 cm	46 cm	20.5 dBi	19.0 dBi	14.5 dBi
n256 8×8	64	55 cm	60 cm	23.0 dBi	21.5 dBi	17.0 dBi

First block n255, second block n256; module = aperture + 2 × 0.2 λ₀ for housing and ground-plane margin.

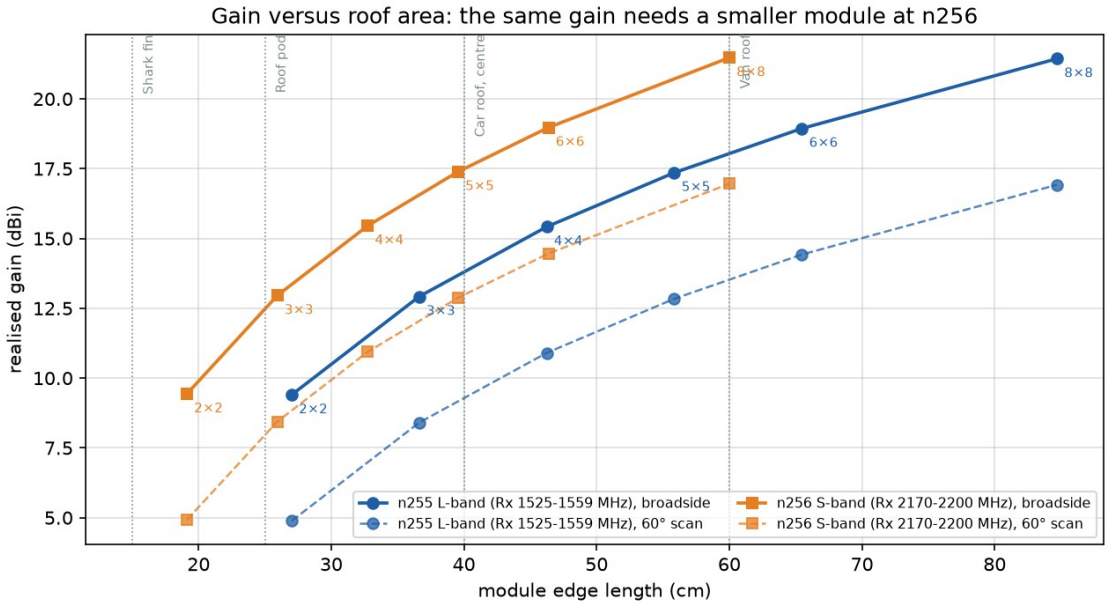


Figure 2. Realised gain versus module edge length, broadside and at 60° scan.

5 Scan and packaging

Electronic steering costs gain in two ways: the projected aperture shrinks with cos θ_0 and the patch element itself rolls off. The model uses $15 \cdot \log10(\cos \theta_0)$, which gives -2.3 dB at 45° and -4.5 dB at 60° (Figure 3).

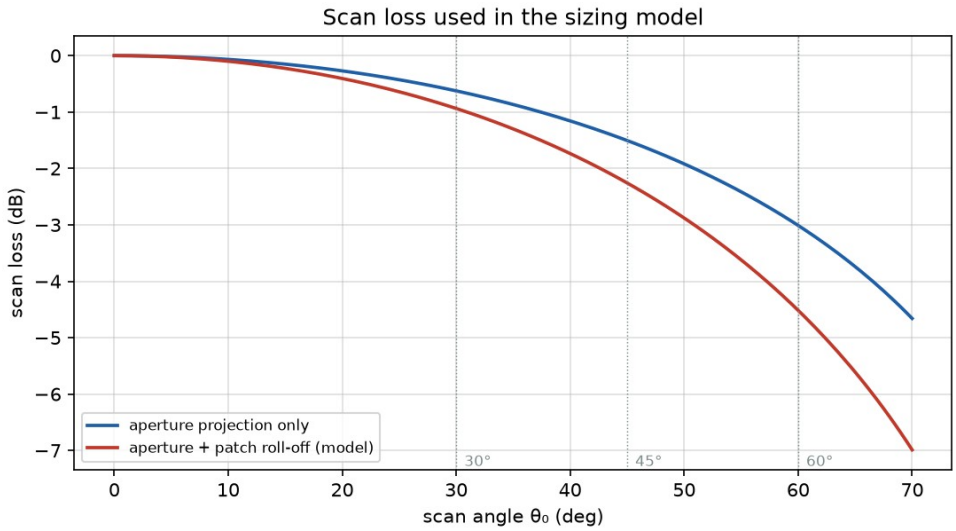


Figure 3. Scan loss model: aperture projection and additional element roll-off.

Figure 4 compares the module edge length with four installation areas; Table 4 lists the largest array that fits each area.

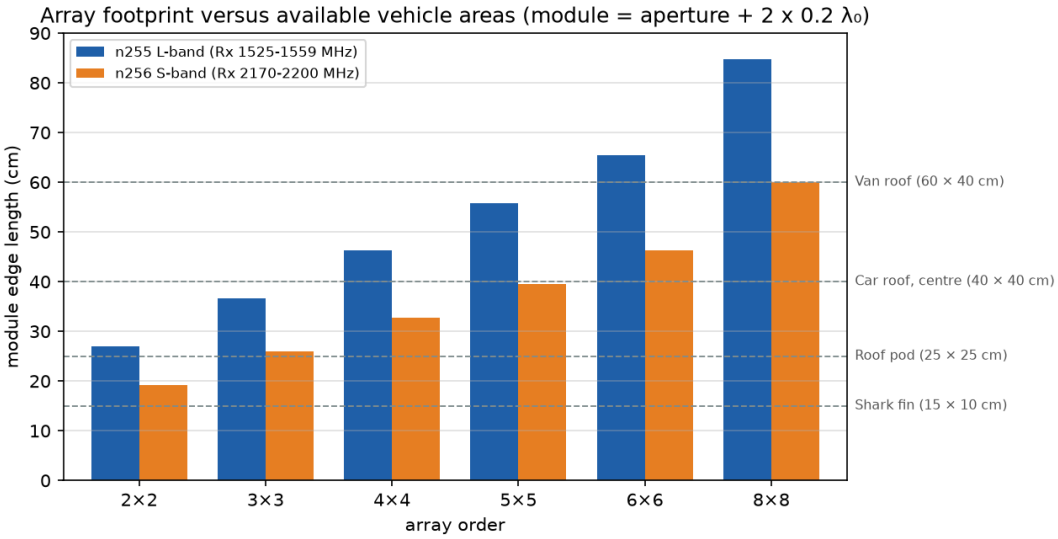


Figure 4. Module edge length versus the installation areas available on a vehicle.

Table 4. Largest array that fits a given installation area.

Area	Size	n255	n256
Shark fin	15 × 10 cm	none	none
Roof pod	25 × 25 cm	none (needs 27 cm)	2×2
Car roof centre	40 × 40 cm	3×3	5×5
Van roof	60 × 40 cm	3×3	5×5

Module = aperture + 2 × 0.2 λ₀; the 40 cm depth of the van roof limits the size more than its width.

6 Link budget

The downlink C/N_0 follows from C/N_0 = EIRP − FSPL + G/T + 228.6 − L_misc, with G/T = G − 10·log10(T_sys). T_sys combines an elevation-dependent antenna temperature (sky + ground pick-up) with a 1.5 dB receiver noise figure. The slant range is computed for a 630 km LEO.

With the placeholder EIRP of 30 dBW the link closes comfortably: a 2×2 reaches 85.9 dBHz at n255 and 82.9 dBHz at n256 at 30° elevation (Figure 5). Even the 85 dBHz broadband-class target is met by 2×2 at n255 and by 3×3 at n256. The packaging size, not the link, is therefore the binding constraint – a conclusion that holds as long as the real EIRP is not far below the placeholder.

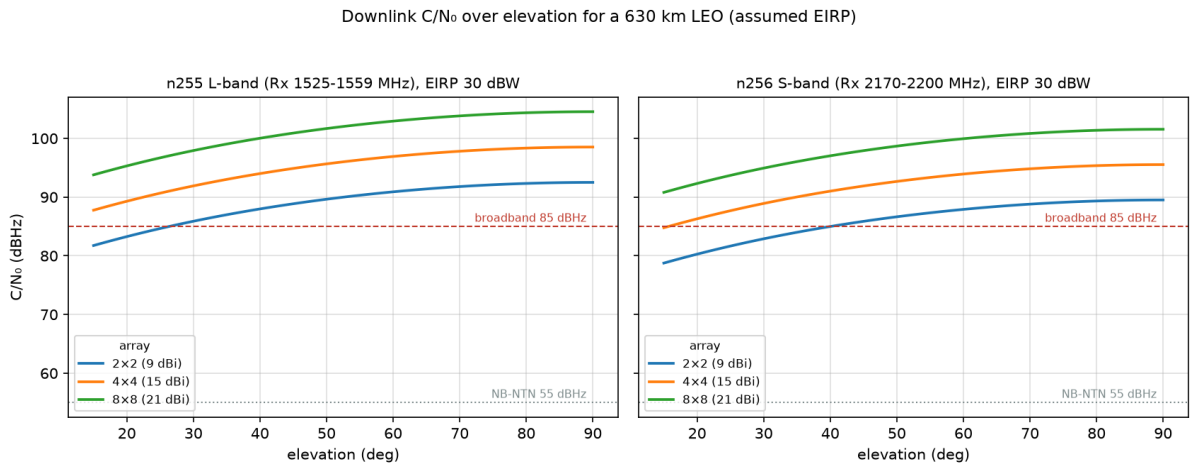


Figure 5. Downlink C/N_0 over elevation for 2×2, 4×4 and 8×8 at both bands (placeholder EIRP).

Figure 6 inverts the question: it shows the minimum array order needed at 30° elevation as a function of the satellite EIRP, for the 70 dBHz and 85 dBHz targets. This is the curve to re-read once the operator EIRP is known.

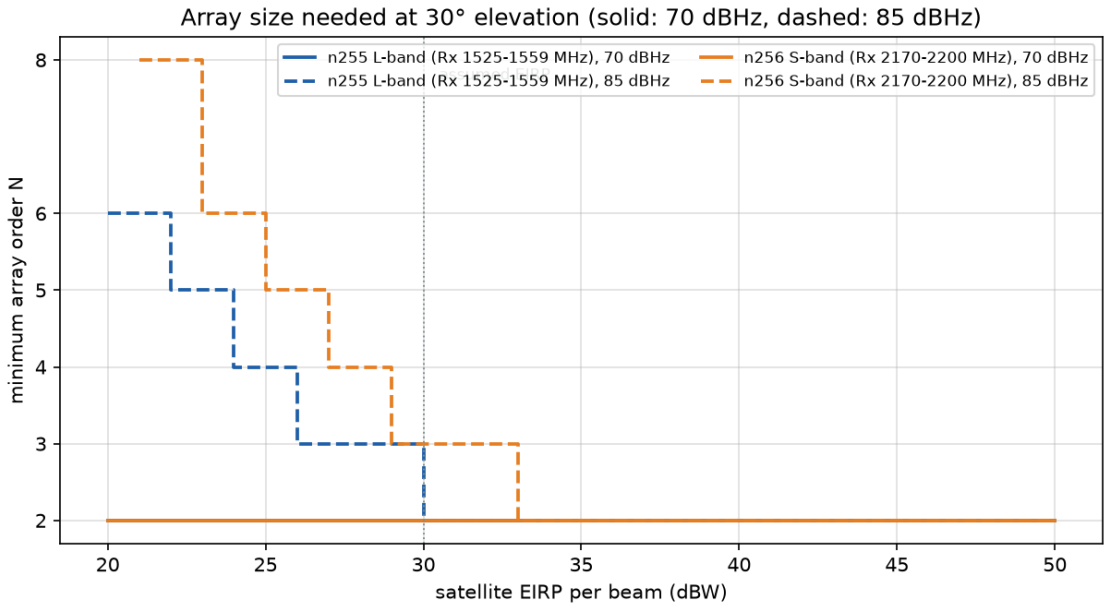


Figure 6. Required array order at 30° elevation versus satellite EIRP.

7 Dual-band and duplex constraints

Two independent constraints prevent a single shared aperture for n255 and n256. First, the frequency ratio is about 1.42, so the n256 λ/2 cell is far smaller than the n255 patch (Figure 7). Second, each band is FDD: together, Tx and Rx span 8.5 % (n255) and 10.5 % (n256) of the band centre, while a single-layer patch delivers 3–5 % impedance bandwidth. Even the Tx/Rx pair within one band therefore needs a broadband (stacked) element or separate apertures.

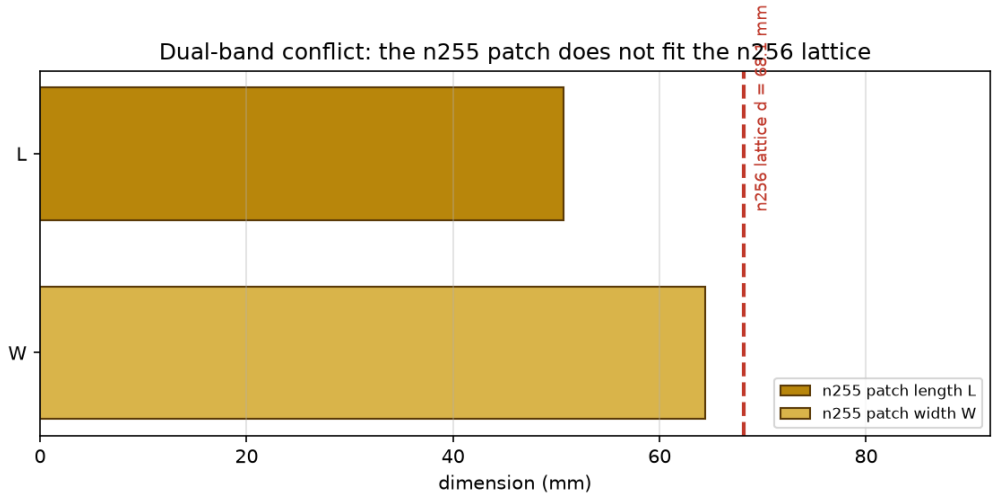


Figure 7. Dual-band conflict: the n255 patch is larger than the n256 lattice cell.

- Two separate modules (n255 + n256) are the simplest and most robust solution.
- A broadband stacked element can cover Tx+Rx of one band but not both bands in one lattice.
- An interleaved shared aperture is possible in principle but doubles the feed complexity and the coupling.

8 Recommendation

For a passenger car the following configuration is the best compromise between gain, roof area and effort:

- n256: 5×5 array, module 40 cm, 17.4 dBi broadside, 12.9 dBi at 60° scan – fits the 40 cm roof centre. In a 25 cm roof pod only the 2×2 (19 cm, 9.4 dBi) fits.
- n255: 3×3 array, module 37 cm, 12.9 dBi – the practical roof-centre limit. The 2×2 (27 cm, 9.4 dBi) needs a pod larger than 25 cm.
- If both bands are mandatory, plan two modules side by side rather than one shared aperture: 37 cm (n255) + 40 cm (n256).
- A mechanical azimuth rotator plus electronic elevation scan reduces the aperture needed for the same link margin and is a valid fallback for a van or truck.

9 Limitations and next steps

- The link budget uses a placeholder EIRP; the required-N curve in Figure 6 must be re-read with operator data.
- The gain is an aperture estimate; mutual coupling, feed-network losses and radome losses need a full-wave run.
- Circular polarisation (RHCP/LHCP) is not modelled; it changes the element design and adds ~0.5 dB.
- The vehicle body acts as a large ground plane; a finite-roof model should quantify pattern and back-lobe changes.
- Next step: full-wave design of the n256 4×4 element → subarray → array chain in the same openEMS workflow as the 77 GHz study.

10 Reproduction

All results are generated by two scripts:

```
python sizing.py # figures 01..07 + docs/results.json
python build_docs.py # this report (EN + DE) + PDF
```

sizing.py contains the model chain and writes every number used in this document to docs/results.json; build_docs.py only formats that file. The model constants (bands, platforms, link assumptions) are in ntn_model.py.